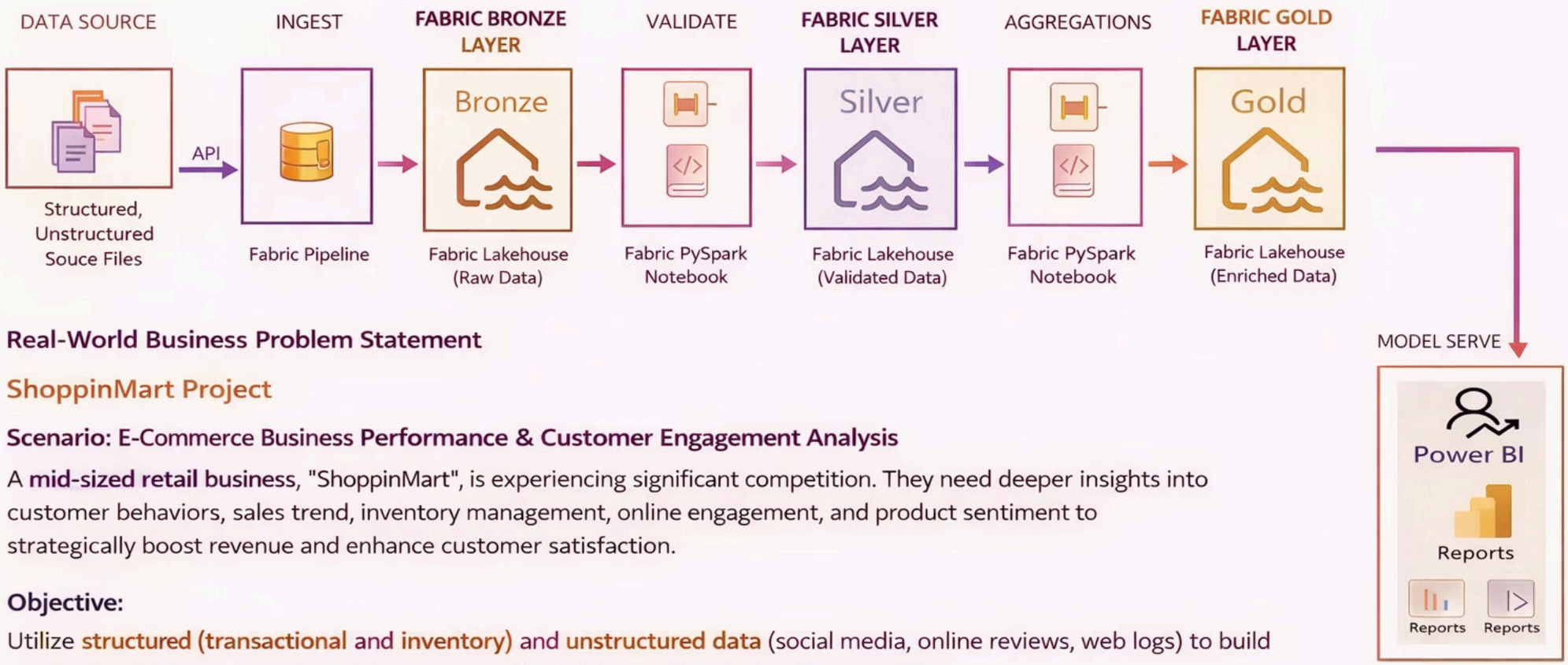
MICROSOFT FABRIC END-TO-END PROJECT (IMPLEMENT MEDALLION ARCHITECTURE)

WITH LAKEHOUSE, DATA PIPELINE, PYSPARK NOTEBOOK, SEMANTIC MODEL and POWER BI



Real-World Business Problem Statement

ShoppinMart Project

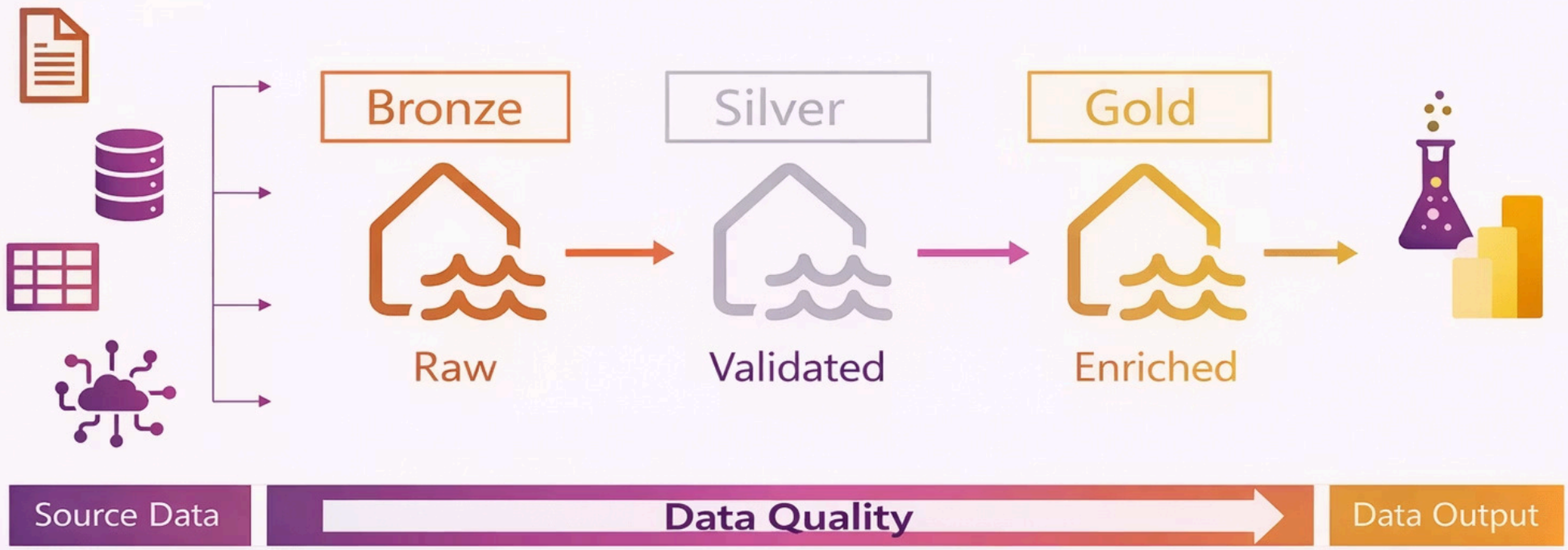
Scenario: E-Commerce Business Performance & Customer Engagement Analysis

A **mid-sized retail business**, "ShoppinMart", is experiencing significant competition. They need deeper insights into customer behaviors, sales trend, inventory management, online engagement, and product sentiment to strategically boost revenue and enhance customer satisfaction.

Objective:

Utilize **structured (transactional and inventory)** and **unstructured data** (social media, online reviews, web logs) to build a unified analytics solution that delivers actionable insights and tracks key business KPIs.

Medallion Architecture



Microsoft Fabric Medallion Key Concepts

BRONZE

- **Raw Zone**
- Stores source data in its original format, including unstructured, semil-structured, or structured data types
- Maintain a source of truth and enable reprocessing and auditing in the future

SILVER

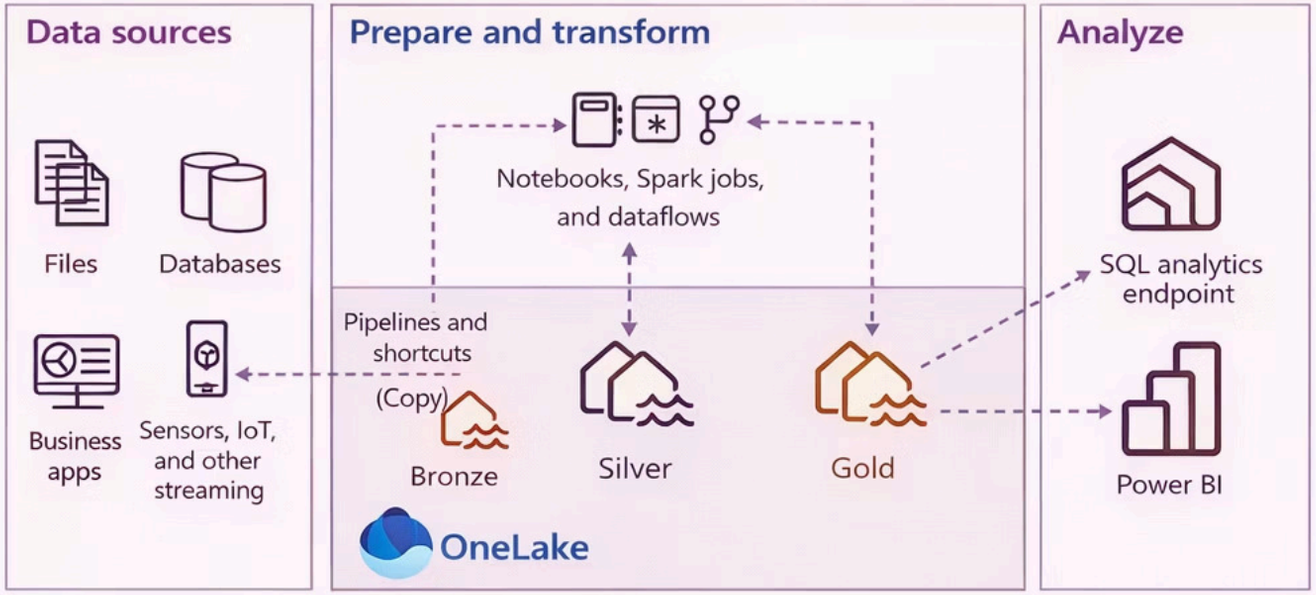
- **Enriched Zone**
- Data is cleansed and standardized
- Validate and Deduplicate your data

GOLD

- **Curated Zone**
- Data is refined to meet specific downstream business and analytics requirements

Medallion Architecture Summary

Layer	Role	Output Format	Purpose
Bronze	Raw Ingestion	CSV, JSON	Preserve original data
Silver	Clean & Integrated Data	Parquet	Accurate analysis-ready datasets
Gold	Aggregated Analytics & KPIs	Parquet	Reporting and dashboards



Source Microsoft

SHOPPING MART ANALYTICS KPI METRICS

Year	Month	Day	Category	Product
2024	All	All	All	All

Top 5 Customers by Sales($)

CustomerName	Sum of TotalAmount
Tammy Smith	$4,959.15
Anita Ruiz	$4,452.02
Susan Diaz	$4,435.02
Linda Massey	$4,757.77
Bradley Norton	$4,329.93

Sales ($) by Category

Total Sales ($)

$254.75K

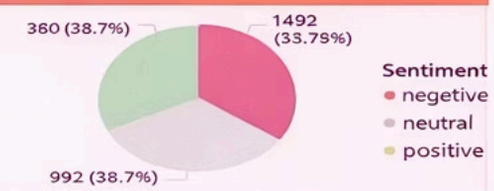
No of Products Sold

2.41K

Top 5 Products by Sales($)

ProductName	Category	Sum of TotalAmount
Chair	Home	$6,118.08
MakeupKit	Fashion	$6,896.78
TreadMill	Electronics	$5,580.98
FitnessWatchBlack	Electronics	$4,757.77
Suit	Fashion	$4,431.14

Social Sentiment



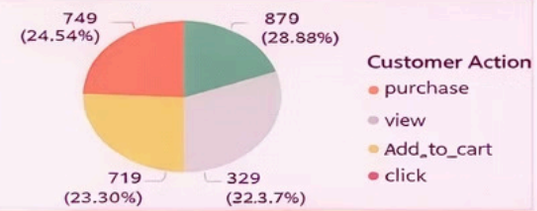
Top 5 Products by Reviews

ProductName	Category	Rating
Suit	Fashion	3.32
MakeupKit	Fashion	3.31
TreadMill	Electronics	3.30
FitnessWatchBlack	Electronics	3.24
		3.24

Top 5 Products by Reviews

ProductName	Rating
Purchase	3.32
New	3.31
Add_to_cart	3.45
Click	3.25
	322

Web Engagement Action



Total Sales by Month